ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 4

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 50

Marking guideline for the Academic Central Grade 4 Natural Sciences and Technology paper 1 · practice b.

Written by Academic Central for practice, following the CAPS curriculum for Grade 4. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words if the science is correct. Do not take marks off for spelling if the meaning is clear.

QUESTION 1

1 1.1 B ✓ 1.2 B ✓ 1.3 A ✓ 1.4 A ✓ 1.5 A ✓ 1.6 B ✓ 1.7 C ✓ 1.8 B ✓ (8)

[8]

QUESTION 2

2 2.1 TRUE ✓ 2.2 FALSE ✓ 2.3 TRUE ✓ 2.4 FALSE ✓ 2.5 TRUE ✓ 2.6 FALSE ✓ (6)

[6]

QUESTION 3

3.1 Cabbage ✓ (1)

3.2 Bird ✓ (1)

3.3 Herbivore ✓. It eats only plants (cabbage leaves) ✓ (2)

3.4 There will be more caterpillars because nothing eats them ✓, so they will eat (destroy) more of the cabbages ✓ (2)

3.5 Any correct chain, e.g. grass → cow → person. A plant first ✓; two animals that really eat what comes before them ✓; arrows point from the food to the eater ✓ (3)

3.6 A lion eats animals such as zebras that eat plants ✓. Without plants these animals would die, and the lion would have no food ✓ (2)

3.7 The maize plant used sunlight to make its food (energy stored in the mealies), and we get this energy when we eat the pap ✓ (1)

[12]

QUESTION 4

4.1.1 Wind (moving air) ✓ (1)

4.1.2 Petrol or diesel (fuel) ✓ (accept electricity for an electric car) (1)

4.1.3 A battery (electricity) ✓ (1)

4.2.1 From Sipho's body (muscles and lungs), which get energy from food ✓ (1)

4.2.2 The air rushes out of the back of the balloon ✓ and pushes the balloon forward, the opposite way ✓ (2)

4.2.3 Hot gases (from burning fuel) ✓ (1)

4.3 The wheel turns on its axle and rolls easily over the ground ✓, so you push the load instead of lifting all of its weight ✓ (2)

4.4 Any ONE: a mattress or bed; a trampoline; a clothes peg; a pen that clicks; a car seat or car; a door that closes by itself; a toy ✓ (1)

[10]

QUESTION 5

5.1	Land (rock and soil) ✓; water ✓; air (atmosphere) ✓	(3)
5.2	It is salty ✓ (it would make us thirstier and sick)	(1)
5.3.1	A ✓	(1)
5.3.2	D ✓	(1)
5.3.3	The Moon does not make its own light; the Sun lights up half of it ✓. As the Moon moves around the Earth, we see different amounts of the lit half ✓	(2)
5.4	Its orbit ✓	(1)
5.5	Any TWO: keeps us warm ✓; dries washing ✓; heats water in solar geysers; helps crops grow and ripen; dries food such as fruit and meat (biltong)	(2)
5.6	The Sun's rays can burn your skin (sunburn) and hurt your eyes ✓	(1)
5.7	Any TWO with a reason: air (oxygen) to breathe, because there is no air on the Moon ✓; water to drink ✓; food to eat; a spacesuit to protect them from the heat and cold	(2)

[14]

TOTAL: 50

END OF MARKING GUIDELINE